Synchronous Mapping and Navigation

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1. Course Content

1. Run the example program to perform mapping and navigation simultaneously in an unknown environment without a grid map.

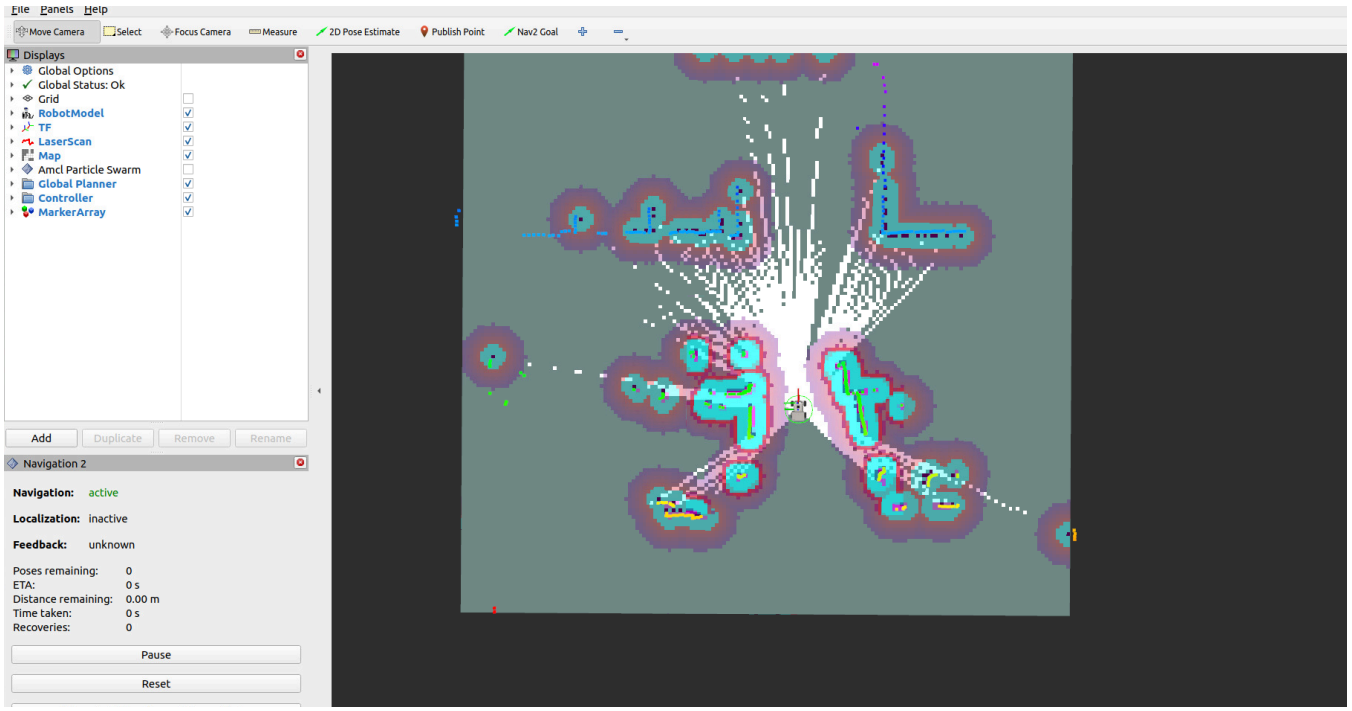
2. Scenario Introduction

- Suitable for scenarios where the map does not need to be saved long-term and the task only needs to be completed once, (after the task is completed, the map can be saved via `map_saver_cli` for later reuse).
- Suitable for scenarios where the robot enters an environment that has not been pre-mapped for the first time; by giving a goal pose, it autonomously builds an environment map and completes traversal of a specified area and arrival at the goal point.

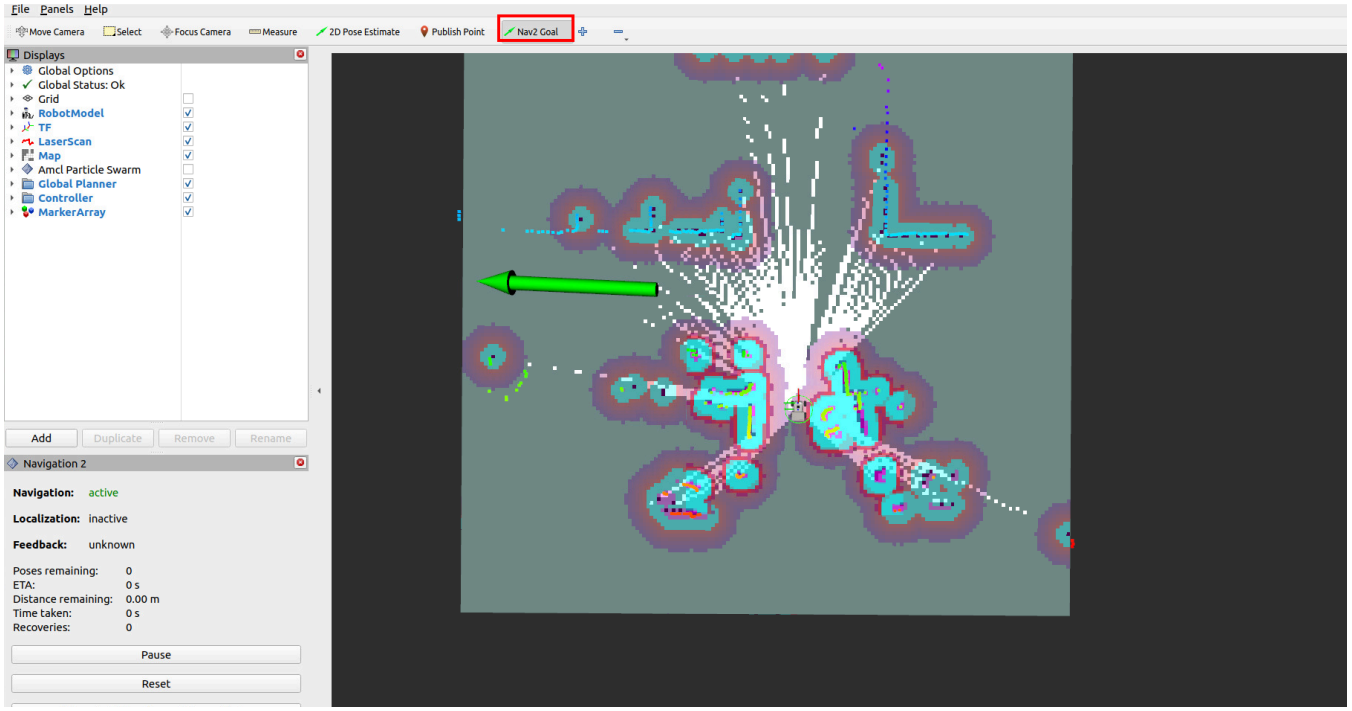
3. Running the Cases

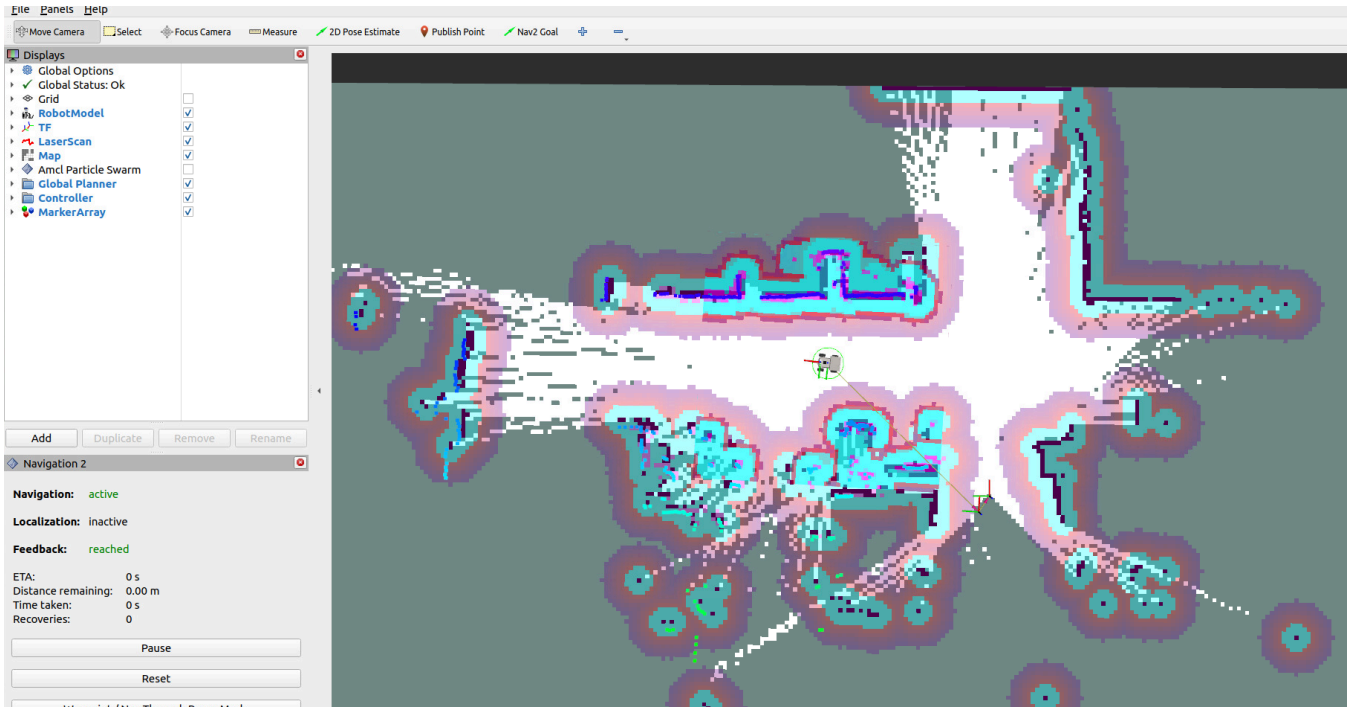
3.1 Starting the Mapping Navigation Node

```
ros2 launch yahboomcar_bringup nav2.launch.py slam:=True
```



- Use the Nav2 Goal tool to give a goal pose, and the robot can then navigate while building the map





3.2 Saving the Map

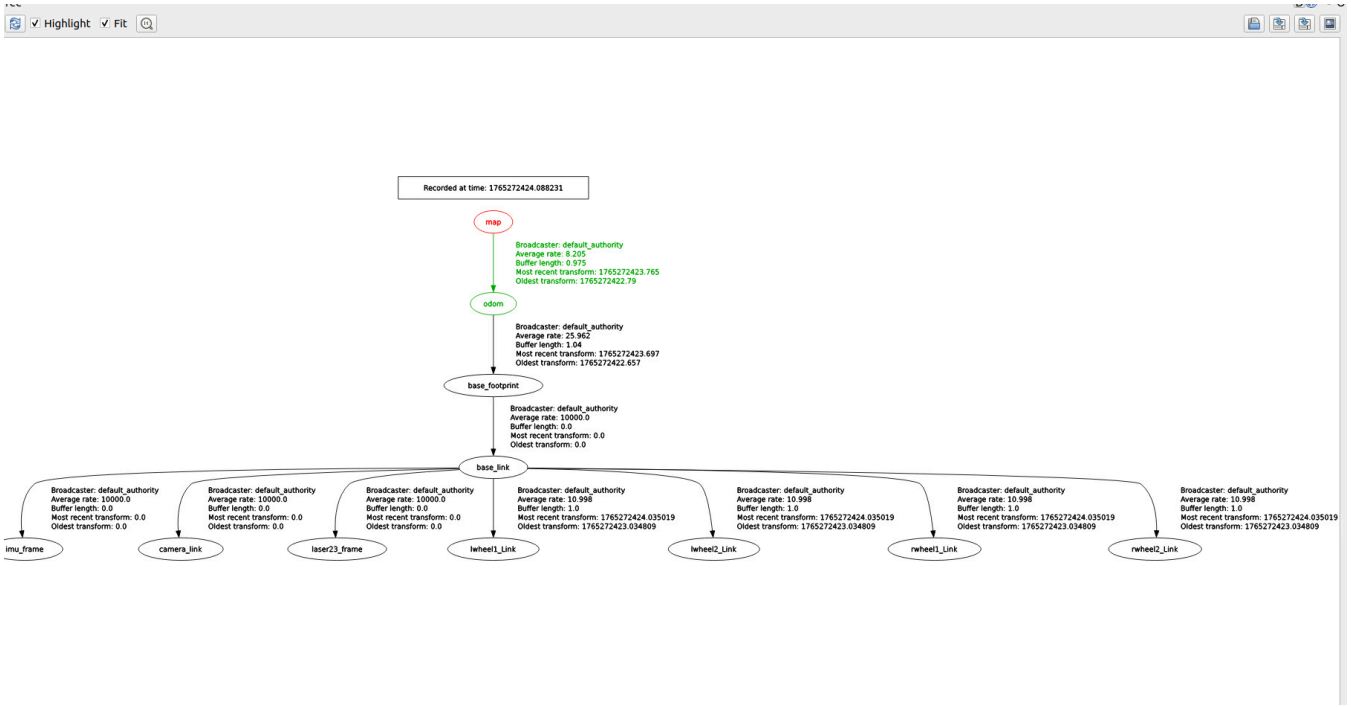
- The commands for saving the grid map and pose map here are exactly the same as in the previous course; you can refer to the map saving method in the prerequisite course [slam-toolbox Algorithm Mapping]

4. Source Code Analysis

4.1 Viewing the TF Tree

```
ros2 run rqt_tf_tree rqt_tf_tree
```

- Here, the TF transform from the map coordinate frame to the odom coordinate frame in the TF tree is provided by slam_toolbox.



4.2 Launch File

- launch file path

```
~/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_bringup/launch/nav2.launch.py
```

- The key program snippet is as follows, which starts the navigation2 navigation stack nodes and enables synchronous mapping mode; the mapping function is provided by the slam_toolbox algorithm.

```
    #Navigate while building the map
    nav_and_slam = GroupAction(
        condition=IfCondition(PythonExpression(["'", localization_mode, "' == 'amcl' and '",
slam, "' == 'True' "])),
        actions=[
            # Navigation using AMCL positioning method
            IncludeLaunchDescription(
                PythonLaunchDescriptionSource(nav2_launch_file),
                launch_arguments={
                    "map": map_file,
                    "use_sim_time": "False",
                    "params_file": params_file,
                    'slam': 'True',
                }.items(),
            )
        ]
    )
```